

Energy Consumption and Efficiency of Biologically Inspired Neural Networks Simulated on GPUs

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Agenda

- Introduction & Problem Statement
- Some Facts about the Problem
- Existing Work on this topic
- Our Approach
- Project Goals
- Project Evaluation Criteria

The Problem Statement

- Artificial Intelligence relies on heavy computing hardware
- Works brilliantly, but uses enormous energy
- Biologically-inspired networks: similar results, far less energy
- Our approach: simulate it on widely accessible hardware; GPUs .

BRIDGING THE GAP: SIMULATING EFFICIENT BIO-INSPIRED AI

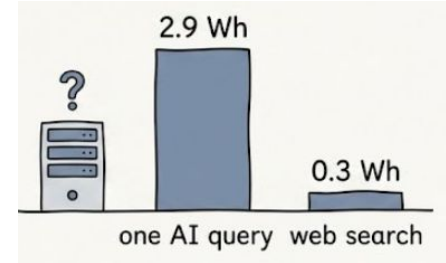


Source: Generated using AI

So how big is this energy problem ?

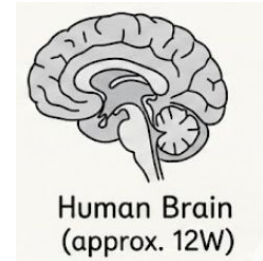
- GPT-3 training $\approx$ **1.3 million kWh**
- one AI query $\approx$ **2.9 Wh vs 0.3 Wh** for a web search

(source : Medium, R. Thompson, Article: AI's Dirty Secret ...)



And why biological inspired approach ?

- nature hints there's a far more efficient way
- **Human Brain** : approx. 12 watts (inspiration, not a current claim)



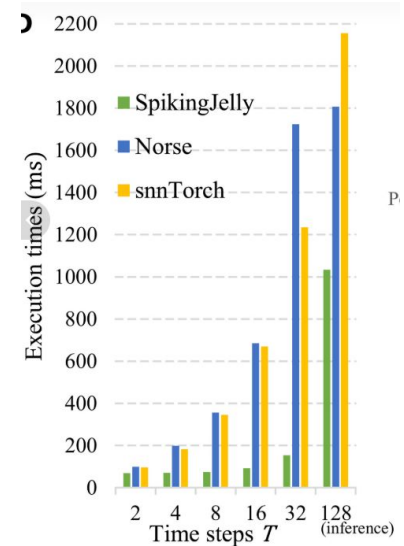
Others' existing work so far:

- Open community project comparing frameworks on raw latency – Open Neuromorphic

(source : <https://github.com/open-neuromorphic/snn-library-benchmarks>)

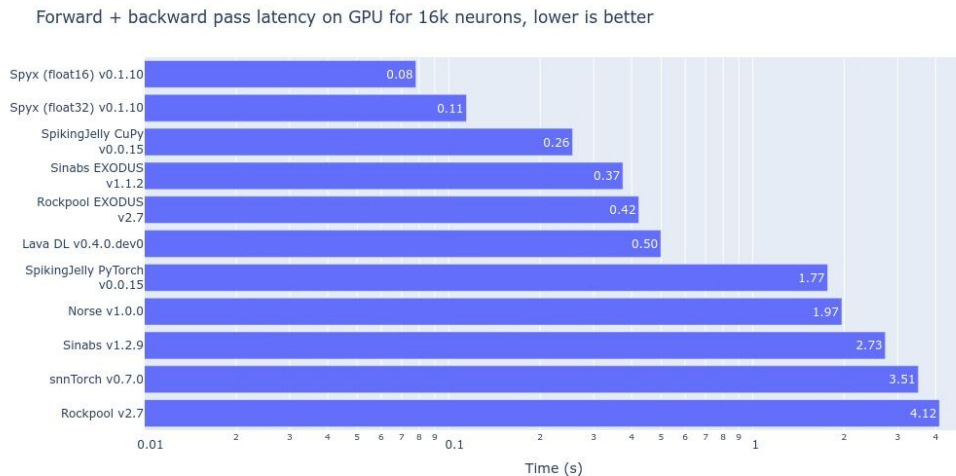
- Compared simulation frameworks, majorly on speed — Science Advances (2023)

(source : <https://www.science.org/doi/10.1126/sciadv.adi1480>)



Source: Model Speed comparisons across frameworks from Science Advances Research Journal

Others' existing work so far (continued)



Our Approach:

- compare accuracy, memory, energy & ease of use together, then optimize further

Our Goal(s)

- Simulate the bio-inspired model on existing hardware (GPUs)
- Compare the simulation frameworks to looking at accuracy, memory, energy - not just speed
- Optimize the simulation pipeline — study effect on energy consumption , memory utilisation

Evaluation Criteria



Memory

GPU memory used during training and inference



Energy consumption

GPU Power/energy consumed for the same task while achieving same accuracy



Performance Accuracy

How correctly the model produces results



Speed

Time to train and to run (per epoch / inference)



Optimization Effect

Change in energy & memory after improving our pipeline



Comparative benchmarking

Comparison of simulation frameworks from use case perspective